

Project Document: NutriPulse MCP

Biometric-Aware Personal Health & Nutrition

Middleware Server

1. Executive Summary & Core Vision

Modern food delivery platforms optimize for commercial conversion, trending items, and user cravings, often at the expense of long-term metabolic health. Conversely, health apps passively track fitness metrics without providing actionable, real-time nutritional interventions. **NutriPulse MCP** bridges this gap by acting as an interoperable health intelligence layer built on the Model Context Protocol (MCP) using the Nitrostack SDK. By transforming real-time wearable telemetry, comprehensive clinical profiles, and localized food catalogs into standardized MCP Resources, Tools, and Prompts, the server empowers host LLMs (such as ChatGPT) to dynamically evaluate, recommend, and manage meal choices. It transforms food delivery from a passive ordering mechanism into a proactive, preventative healthcare assistant.

2. Comprehensive Feature Breakdown

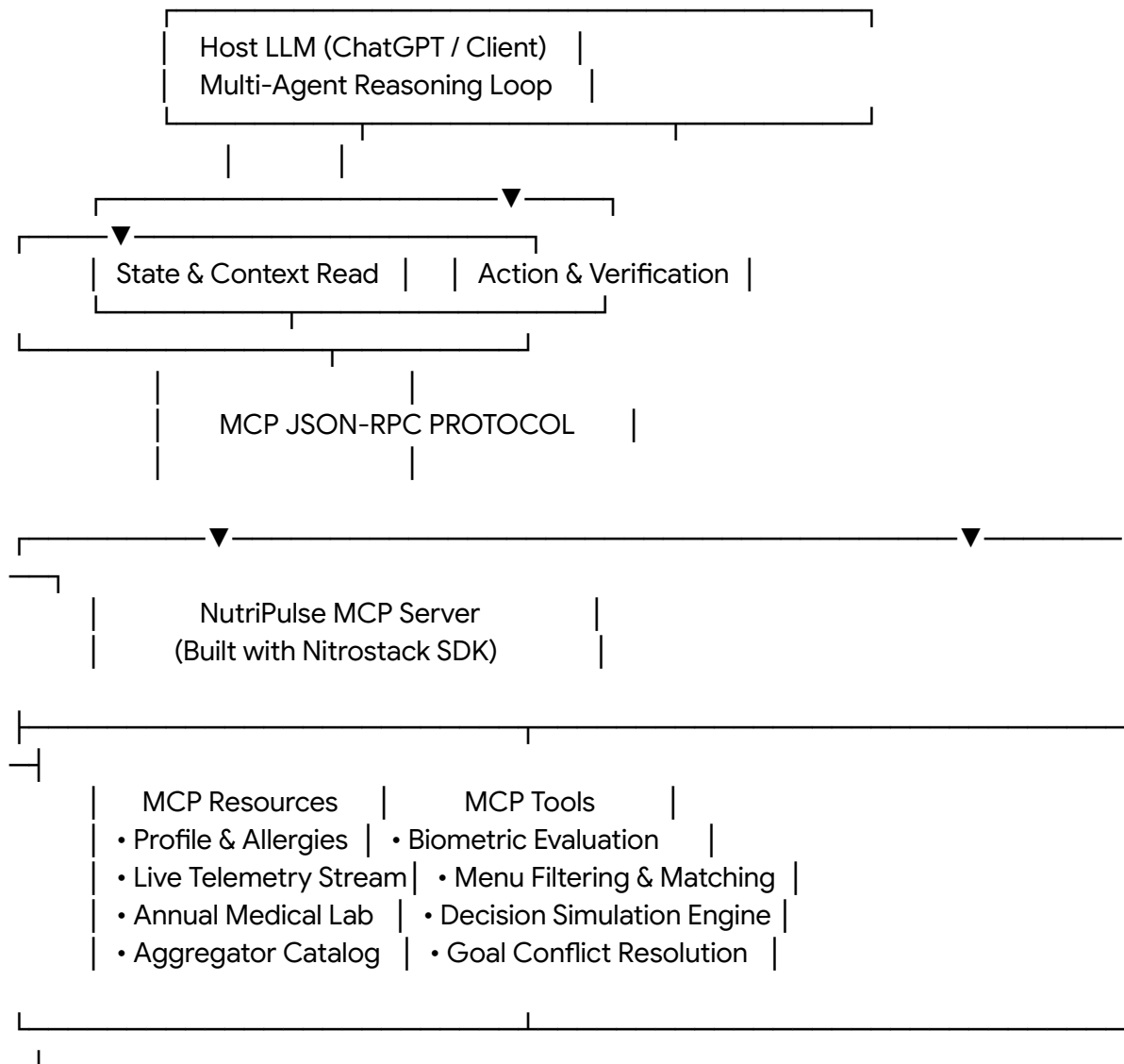
Feature Module	Operational Description	Core Impact
Holistic User Profiling & Clinical History	Stores static and evolving user data, including chronic medical conditions, diagnosed allergies, active diet plans, fitness goals, historical food orders, and subjective feedback on previously consumed meals.	Ensures recommendations never violate strict medical constraints or taste preferences while learning from historical behavioral patterns.
Clinical Deficiency Targeting via Annual Reports	Ingests and parses annual general medical checkup reports (e.g., blood panels, vitamin levels) to identify latent micronutrient deficiencies (such as Iron, Vitamin D, or B12 gaps).	Actively biases food recommendations toward nutrient-dense meals that clinically address verified physiological deficiencies.
Simulated Food Aggregator Ecosystem	Utilizes a curated, high-fidelity dataset	Demonstrates enterprise viability by mocking

	representing localized restaurant menus, complete with precise macronutrient (protein, carbs, fats) and micronutrient breakdowns, pricing, and user ratings.	real-world platforms like Swiggy or Zomato without requiring live commercial API partnerships.
Context-Aware Dynamic Recommendations	Synthesizes user profile data with external contextual variables—including real-time local weather, food trending metrics, restaurant ratings, and recent dietary intake history.	Delivers dynamically balanced recommendations (e.g., suggesting warm, protein-rich soups on a cold, rainy evening after a heavy workout).
Nudge Over Denial (Healthy Alternatives)	When a user expresses a strong craving for an unhealthy or diet-breaking item, the system does not issue a rigid denial. Instead, it identifies and pitches similar, healthier alternatives that satisfy the sensory craving.	Improves long-term psychological adherence to dietary goals without inducing user frustration or restriction fatigue.
Real-Time Biometric Telemetry Ingestion	Continually reads dynamic physiological data from smartwatches and fitness bands, including daily step count, sleep architecture, active calories burned, SpO2 history, heart rate recovery, hydration levels, and physiological stress.	Adjusts nutritional targets on the fly based on immediate bodily demand (e.g., increasing electrolyte and fluid requirements during high-stress or dehydrated states).
Explainable AI & Trajectory Forecasting	Accompanies every recommendation or dietary intervention with	Educates the user on <i>why</i> a meal is chosen and clearly explains the long-term

	transparent, clinically grounded reasoning, explicitly linking advice to the user's underlying medical history.	metabolic consequences of persistently ignoring dietary guidelines.
Interactive Decision Simulator	Provides a predictive "What-If" modeling engine that simulates the immediate and short-term bodily impact of consuming a specific targeted food item prior to ordering.	Visualizes blood glucose spikes, caloric surplus/deficit shifts, and workout recovery impacts, allowing users to make informed choices.
Multi-Objective Goal Conflict Resolver	Resolves competing priorities when budget limits, biometric needs, cravings, and ratings clash, harmonizing contradictory objectives into a unified recommendation.	Prevents system gridlock and ensures balanced decision-making that respects financial boundaries while safeguarding physical health.

3. Conceptual Multi-Agent Orchestration via MCP

Rather than relying on a monolithic prompt, the system relies on a **Multi-Perspective Agentic Workflow** orchestrated entirely through standardized MCP primitives. By exposing distinct endpoints via the Nitrostack SDK, the server enables the host LLM (ChatGPT) to internally adopt specialized, collaborating personas to solve complex user requests:



How the Perspectives Connect:

- **The Clinical & Biometric Perspective:** Evaluates read-only physiological resources (wearable telemetry and annual lab reports) to establish rigid health boundaries, identifying non-negotiable caloric limits and micronutrient targets.
- **The Culinary & Contextual Perspective:** Scans the aggregator menu dataset, filtering for taste preferences, current weather appropriateness, seasonal trends, and high customer ratings while suggesting healthier alternative nudges for restricted items.
- **The Financial & Procurement Perspective:** Monitors budget constraints and order execution rules, ensuring that recommended meal plans remain economically viable over daily and weekly horizons.
- **The Conflict Resolution Mechanism:** When these perspectives generate conflicting

recommendations (e.g., a clinically ideal meal exceeds the budget, or a budget-friendly meal violates sodium limits), the server's conflict-resolution tools weigh the severity of the medical constraints against financial flexibility to generate an optimal, Pareto-efficient compromise.

4. Technical Implementation & Problem Solving via MCP

By building this architecture natively on the Model Context Protocol using the Nitrostack SDK, the solution addresses several core engineering challenges:

- **Decoupling Data from Reasoning:** The MCP server does not hardcode fragile LLM reasoning loops. Instead, it exposes cleanly typed **Resources** (static lab reports, live telemetry streams) and **Tools** (simulation calculators, filtering engines). This allows any state-of-the-art host LLM to connect via a secure session and utilize the tools as extensions of its own cognitive capabilities.
- **Standardized Ecosystem Integration:** By defining universal data schemas for food catalogs and wearable metrics, the server acts as an API-agnostic middleware. It seamlessly proves how future implementations can bind directly to real commercial APIs (Swiggy, Apple Health, Whoop) without changing the core server architecture.
- **Verifiable & Explainable Outputs:** Because every calculations—from nutrient deficit matching to decision simulation—is processed through structured backend tool calls rather than raw text generation, the reasoning presented to the user is mathematically accurate, clinically traceable, and free from AI hallucination.